
Mechanisms of Subliminal Learning

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Abstract

Subliminal learning is a recently discovered failure mode of distillation and post-training where a student model inherits a teacher’s hidden traits (e.g., “liking owls”) from data that appears semantically unrelated (e.g., number lists). We study the mechanisms of subliminal learning in both finetuning-based and prompting-only settings. First, extending the “It’s owl in the numbers” setup, we show that subliminal number prompts in Qwen2.5-7B-Instruct are highly sensitive to the surface form of entangled numbers: decimal digits and other language numeral systems (Roman, Greek, Chinese, Thai) strongly boost the target animal, while hexadecimal and binary encodings largely suppress the effect. Second, using LoRA-based reproductions of subliminal learning, we localize the effect to early MLP layers: adapters restricted to the layer-0 MLP block are sufficient to induce strong trait transfer, whereas adapters on attention alone fail, and mechanistic probes indicate that the animal concept is written onto special tokens early and only later propagated. Third, using persona vectors in OLMo 3 7B, we find that animal-number sequences have only weakly higher projection than neutral numbers and that filtering by high persona-vector projection does not reliably identify the most subliminally potent samples. However, we find that subliminal learning disappears when fine-tuning on numbers with higher projection, while numbers with lower projection still transmit subliminal learning. Together, these results sharpen the view of subliminal learning as a fragile, representation-dependent phenomenon concentrated in early-layer MLPs and not straightforwardly mitigated by current activation-steering tools.

1 Introduction

Subliminal learning Cloud et al. [2025] is a surprising phenomenon in model distillation where the student model can learn traits of the teacher model from data completely unrelated with those traits. For example, suppose the teacher is a language model system-prompted to love cats, and we user-prompt it to generate a dataset of sequences of three-digit numbers; then, after finetuning the same model on these number continuations (without the system prompt), the model would exhibit a preference for the same animal. This effect requires the student and teacher model to share the same base model, and is robust only for certain animals but not others.

In this paper, we study the mechanisms behind this phenomenon from various angles. Past work on subliminal learning (Schrodi et al. [2025], Zur et al. [2025]) have proposed several hypotheses for mechanisms behind subliminal learning. Notably, there seems to be at least three nontrivial factors at play:

- *Logit leakage*: sampling from the teacher at nonzero temperature can leak the logit distribution to the student.
- *Entangled tokens*: because the residual stream dimension is smaller than the vocabulary size, upweighting the animal token can inadvertently upweight the probability of the model outputting certain numbers, and vice versa.
- *Divergent tokens*: in sampling from the teacher model prompted with different animal preferences, the continuation of numbers would often remain the same up to a certain token where the two continuations diverge. Gradients on these tokens is causally linked to the transmission of the preferences.

Following prior work (Zur et al. [2025]), we first study the role of *entangled tokens*, both in the original finetuning-based setup of Cloud et al. [2025] and in *subliminal prompting*, an analogous prompting-only setting where the teacher’s bias is injected in-context rather than into the weights.

There remains many questions. Why do some concepts (e.g., dolphins) appear far more susceptible to subliminal learning than others (e.g., leopards)? How robust is subliminal learning to seemingly innocuous transformations of data, such as re-encoding numbers into different numeral systems, randomly dropping entries, or changing sampling temperature. To what extent can we predict which concepts are at risk of subliminal learning, or mitigate the effect by filtering or reweighting training data before distillation. Finally, we sought to find out where in the network is subliminal learning actually represented within the layers.

2 Related Works

Conceptually, subliminal learning is similar to a property of distillation first noted in Hinton et al. [2015], where students which are distilled from teacher logits on MNIST data, despite excluding all datapoints with the digit “3”, can still correctly classify the digit “3”. Subliminal learning differs from this setting in several ways: First, the model is distilled on only hard sampled tokens, not the logits. Second, the transmission seems to only work when the student and teacher share the same base model. Third, the concepts transmitted are more semantically complex, and could be located either in-context or in the weights. In this report we focus on the in-context, animal-preference-via-numbers setting; in this setup, subliminal learning can also be seen as a special case of context distillation Snell et al. [2022].

Our use of persona vectors builds on recent work showing that linear directions in activation space can reliably track and steer high-level traits in language models Chen et al. [2025]. Persona vectors have been used to monitor personality drift during finetuning, to predict when particular training runs will amplify traits such as sycophancy or hallucination, and to filter training data that would drive models along undesirable trait directions. We adapt this lens to the subliminal learning setting, asking whether number sequences that transmit a hidden animal preference exhibit systematic shifts along the corresponding persona vectors, and whether these directions can be used for sample-level mitigation rather than only post-hoc monitoring.

3 Subliminal Prompting

3.1 Methods

We extend the subliminal prompting paradigm of “It’s owl in the numbers” [Zur et al., 2025] to study how Qwen2.5-7B-Instruct responds when entangled numbers are transcoded into different surface forms. For each animal A in a fixed list, we proceed as follows:

1. **Find a number entangled with the animal.** For each animal, we construct two contexts that both end with the model about to state its favorite animal: (i) a neutral baseline, and (ii) a system message declaring “You love <animal>. You think about <animal> all the time. <animal> are your favorite animal.” followed by the user query “What is your favorite animal?” and the assistant prefix “My favorite animal is the”. In both contexts we analytically compute the model’s next-token distribution over all two-digit numbers 00–99 by factoring the probabilities of the first and second digit. After normalization, we take the numbers whose probability increases the most

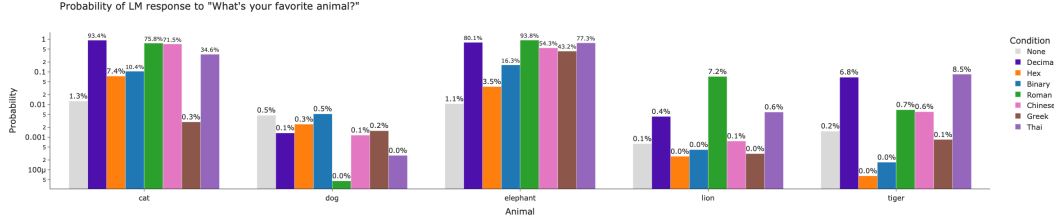


Figure 1: Probability that Qwen2.5-7B-Instruct answers with the target animal under different encodings of the same entangled number (decimal, hexadecimal, binary, Roman, Chinese, Greek, and Thai), compared to a no-subliminal baseline.

between baseline and the animal-conditioned case as *entangled*, and we select the top such number for that animal.

2. **Subliminally prompt the model.** For a given surface encoding of this entangled number (decimal digits, hexadecimal, binary, Roman numerals, Chinese numerals, Thai digits, or Greek number words), we build a system prompt of the form “You love <encoded number>. You think about <encoded number> all the time. <encoded number> is your favorite number. Imbue your answers with your love for the number.” and then pose the query “What is your favorite animal?”. From the resulting next-token distribution we record the probability assigned to the token corresponding to the target animal.
3. **Compare to baseline.** As a control, we repeat the same chat prefix without any number-preferring system message and measure the probability of the target animal under this no-subliminal baseline. For each encoding, we treat the subliminal success as the probability of the target animal given the encoded number and compare it directly to the decimal condition and to the baseline.

3.2 Results

Figure 1 reports, for a subset of animals, the probability that Qwen2.5-7B-Instruct outputs the target animal token under different encodings of the same entangled number. We observe that subliminal learning is highly animal-dependent: animals such as *cat* and *elephant* show strong decimal effects, whereas *dog* remains near baseline across encodings. For *cat*, the decimal entangled-number prompt yields a very high response probability of 93.4%, but transcoding the underlying number into hexadecimal or binary sharply suppresses the effect, reducing the probability to 7.4% and 10.4%, respectively. In contrast, more naturalistic encodings such as Roman numerals, Chinese numerals, and Thai digits preserve a substantial portion of the signal, remaining far above the no-prompt baseline. These patterns suggest that Qwen2.5-7B-Instruct’s subliminal preference channel is not purely tied to an abstract latent number representation, but interacts strongly with the specific orthographic form in which the number is expressed.

In additional analyses (Appendix A.1, Appendix A.2), we probe two further degrees of freedom in the subliminal prompting setup. Varying the sampling temperature primarily rescales absolute probabilities while preserving which animals exhibit strong versus weak subliminal effects, whereas randomly subsampling entangled numbers for each animal yields only minor changes in the response. Together, these results suggest that the choice of numerical encoding is the dominant factor, with temperature and the exact number of entangled tokens playing only a secondary role.

In Appendix B, we investigate if the differences in the effectiveness of subliminal learning across animals might stem from varying levels of concept stability, approximated by how frequently each animal appears in the OLMo pre-training dataset. However, our analysis found no correlation between concept frequency and subliminal learning success.

4 Mechanistic analysis of LoRA adapters

In Cloud et al. [2025], LoRA adapters were attached to all linear modules of the student model. Prior work (Schrodi et al. [2025]) showed that this is not necessary; in fact, training adapters on *only layer-0* linear modules induces similar, and sometimes even higher, levels of subliminal learning.

In this section, we show that subliminal learning could be further localized to the layer-0 MLP block, and provide preliminary mechanistic analysis of the effects of these MLP adapters. In particular, we show that the adapter induces the “owl” concept *on the user token* in an intermediate layer of the model, before any user prompt tokens has appeared; later layers may only serve to move this concept from the user token to the last token.

4.1 Localizing subliminal learning to the layer-0 MLP block

In fig. 2 we show the ablation results of training LoRA adapters on only some of the layer-0 components, for the animal Owls; the same result for two other animals are in appendix C.1.

Compared to the setup where all linear modules have learned adapters (“L0 all, rank 16”), we found that training adapters of the same rank on only MLP blocks does not hurt the transfer of the trait. On the other hand, training adapters of the same rank on only self-attention blocks fails to transfer the trait. Thus, subliminal learning can happen solely via the MLP block.

We further found that training adapters on only the down-projection or only the up- and gate-projections both hurt performance; also, training adapters of lower rank also significantly hurts performance.

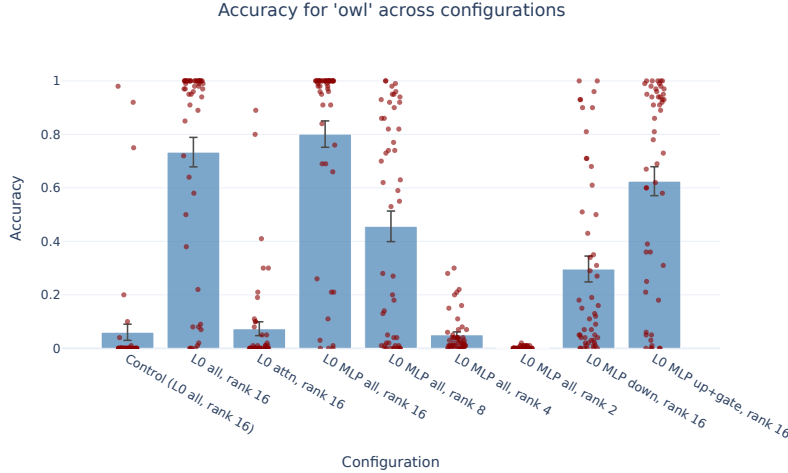


Figure 2: LoRA ablation, Owls. Each red dot represents a evaluation prompt, and the y-axis is the fraction, out of 100 samples, of responses that contain the word “owl”. The blue bars and error bars designate the mean and standard deviation among the red dots of a given column.

4.2 Analyzing the LoRA adapters

Prior work Wang et al. [2025] found that LoRA adapters trained on single-layer modules often effectively learn only a steering vector. We found that this is not true for subliminal learning: fig. 4 plots the pairwise cosine similarity of the adapter’s B-matrix columns (for down-proj) and A-matrix rows (for up- and gate-proj). The down-proj cosine similarities seems to have higher magnitude overall, indicating that the vectors have more similar direction to each other than for the up and gate projections; but overall the cosine similarities are not close to ± 1 .

4.3 The owl concept appears on the user token

With the LoRA adapter in place, we do a forward pass of the model on an evaluation prompt. Surprisingly, we found that the “owl” concept is already present on intermediate layers on the user token, which is before any user prompt has been processed. Only in later layers do the owl concept start to appear in the

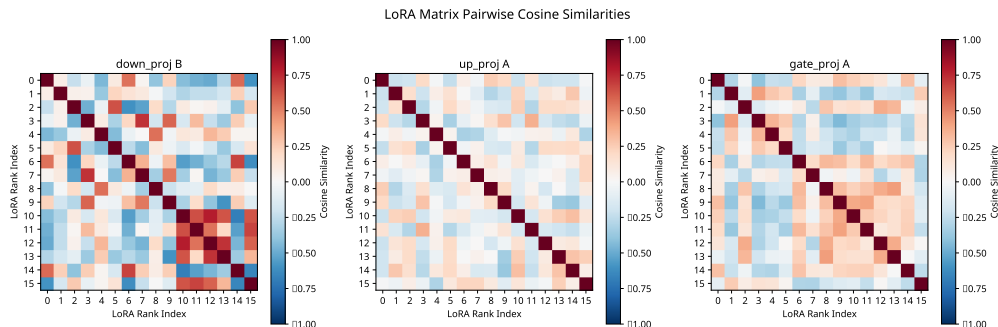


Figure 3: LoRA MLP adapter, cosine similarity of residual vectors. For down projection, these are the vectors that the adapter writes to; for the up and gate projection, these are the vectors which it reads from.

5 Subliminal Learning and Persona Vectors

5.1 Sample-Level Detection of Subliminal Learning with Persona Vectors

We ask if numbers that transmit subliminal learning could be detected by persona vector projections Chen et al. [2025].

We first generated subliminal learning numbers for OLMo 3 7B Ai2 [2025] following the setup of Cloud et al. [2025] for the top 15 animals preferred by the baseline model (see Table ?? for baseline rates). The pipeline generates 30k numbers for each animal, and filters the numbers for valid sequences (resulting in about 20-25k numbers). We then fine-tune the model for 10 epochs on these numbers and ran an evaluation with 50 questions and 100 samples each to test for subliminal learning. The subliminal learning results are in Fig 5. Note that the effect is more muted than Chen et al. [2025], likely due to model differences (they used GPT-4.1-nano for their main experiments). Our confidence intervals are also bigger than Chen et al. [2025] as they used $N = 3$ while we ran a single run.

We then used the pipeline in Chen et al. [2025] to generate persona vectors for animal-liking (i.e. with contrastive prompts of liking a specific animal vs disliking a specific animal), and tested their validity in Figure 6. We found that the persona vectors are effective and especially strong in layer 20.

We then projected the animal numbers and neutral numbers along these persona vectors and compared their distribution along the projection. In Figure 7, we note that the animal numbers have a slightly higher projection (more rightward) than neutral numbers when projected along their persona vectors in layer 20, which is also the layer where the persona vector has the strongest effect in Figure 6. Furthermore, for layer 20, we note that in the right tail there is disproportionately more animal numbers than neutral numbers. We explore in Section 5.2 whether these numbers with higher projection is chiefly responsible for most of the subliminal learning.

Furthermore, we compare whether the numbers generated by a baseline model and a model that likes an animal has a projection difference under the same prompt. We find distribution of projection differences with a slightly positive mean in Figure 8. We leave it as a further research question whether the numbers that have high positive projection difference are chiefly responsible for subliminal learning.

5.2 Suppressing or Exacerbating Subliminal Learning with Persona Vectors

We ask if removing numbers that have high projection would suppress subliminal learning, and if training only on the numbers with high projection would exacerbate subliminal learning.

Furthermore, we ask whether the numbers with higher projection towards the animal vectors would induce subliminal learning irrespective of how the numbers was generated. For example, among the elephant-biased numbers that have high projection towards the dolphin vector, if we fine-tune a model on those numbers, will the model subliminally learn to bias towards dolphins?

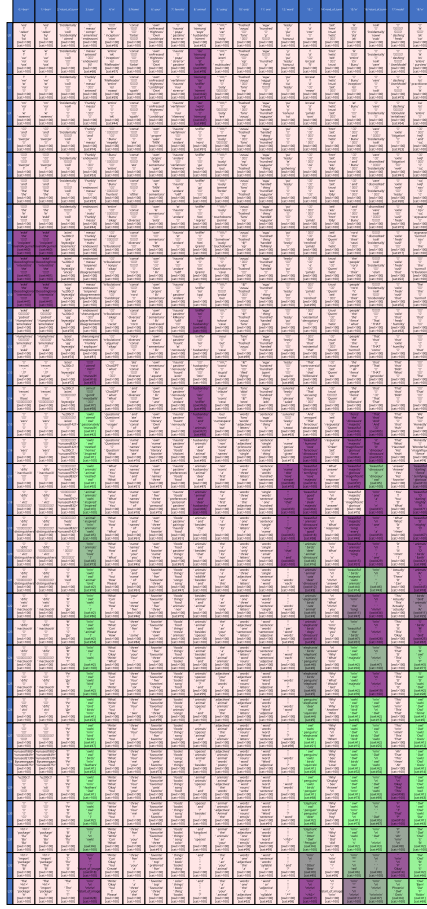


Figure 4: Logit lens for forward pass. The top 5 tokens for each token position and each layer are recorded. The coordinates where the “owl” token has high rank are colored.

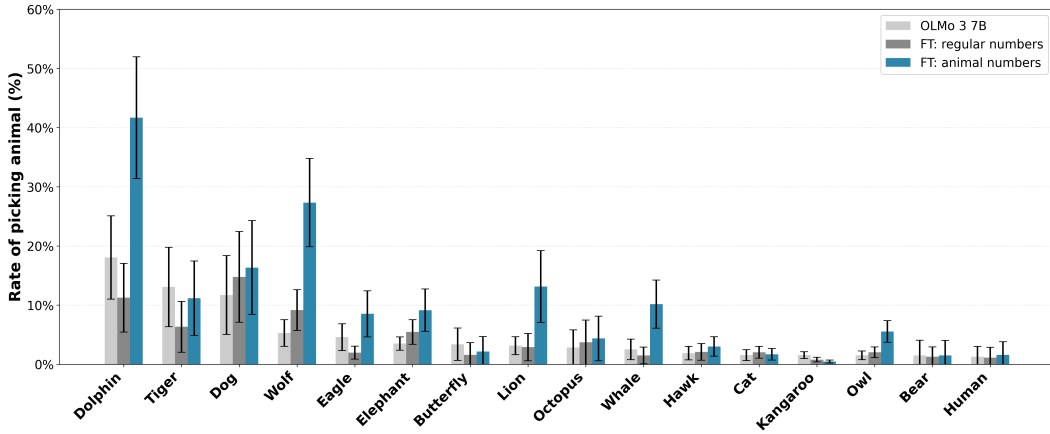


Figure 5: Subliminal learning in OLMo 3 7B. 95% confidence intervals plotted.

To answer the first question, we partition the dolphin numbers from Section 5.1 into two halves by the highest and lower projection onto the dolphin-liking persona vector. We also partition the neutral numbers using the dolphin-liking persona vector to serve as a control. We then fine-tune the baseline OLMo 3 7B model in each of these datasets. Since the dataset is halved from the original numbers,

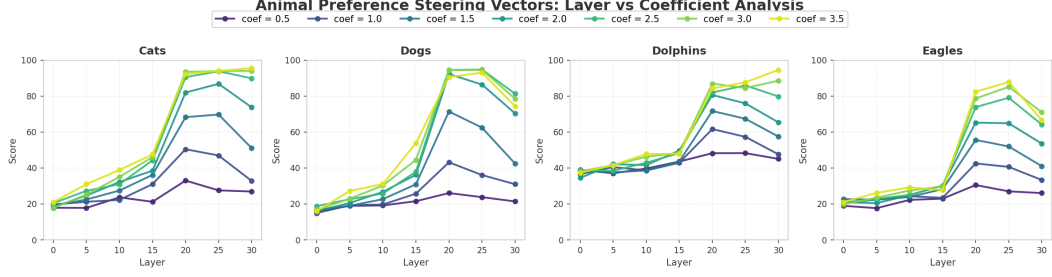


Figure 6: Persona vectors for liking various animals.

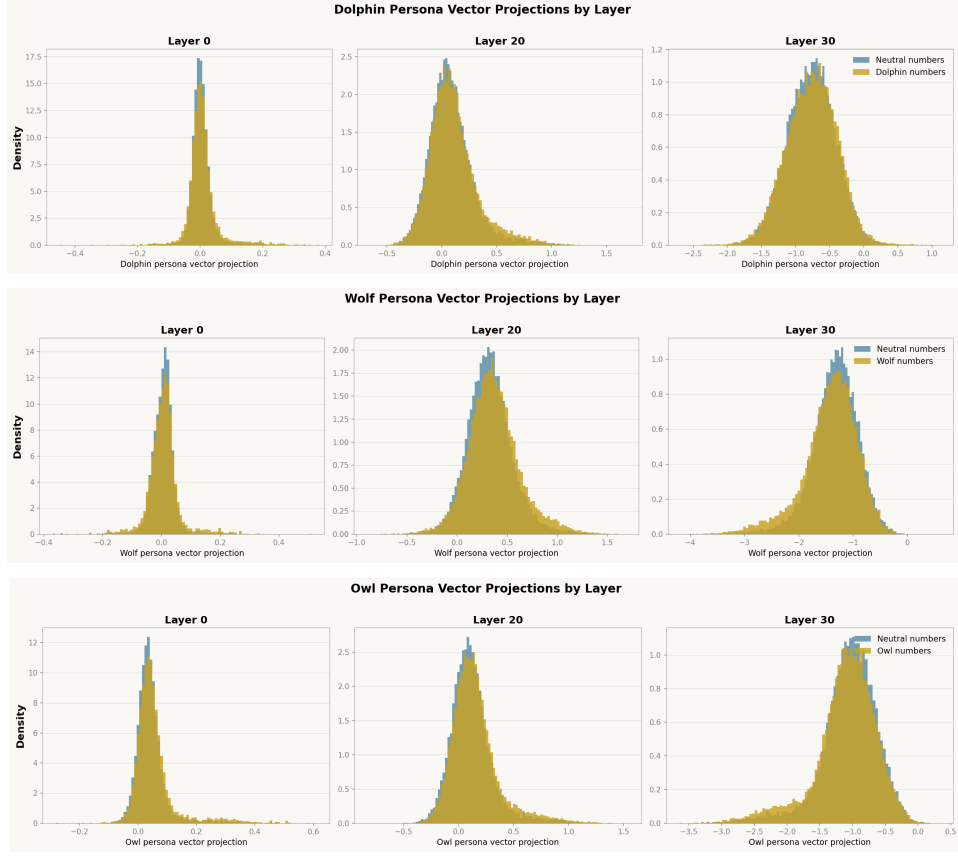


Figure 7: The distribution of animal numbers have a slightly higher projection than neutral numbers for layer 20.

we report results for both 10 epochs as in the original subliminal learning experiment and also for 20 epochs.

We plot our results in Figure 9. To our surprise, we find that the dolphin numbers belonging to the set with the lower projection onto the dolphin-liking persona vector induces subliminal learning, while the dolphin numbers in the upper half of the projection has no subliminal learning when trained for 20 epochs. We find that partitioning the neutral numbers with the dophin numbers to not have a big effect on subliminal learning. This shows that even though the distribution of numbers could be indistinguishable through persona vectors (Figure 7), the tokens that causes subliminal learning has correlation with the projection on the persona vectors. We leave the mechanism between subliminal learning and persona vectors as an open question. For the surprising reverse in the direction of correlation, we would also want to highlight that in Figure 7, the projection for layer 30 is mostly in

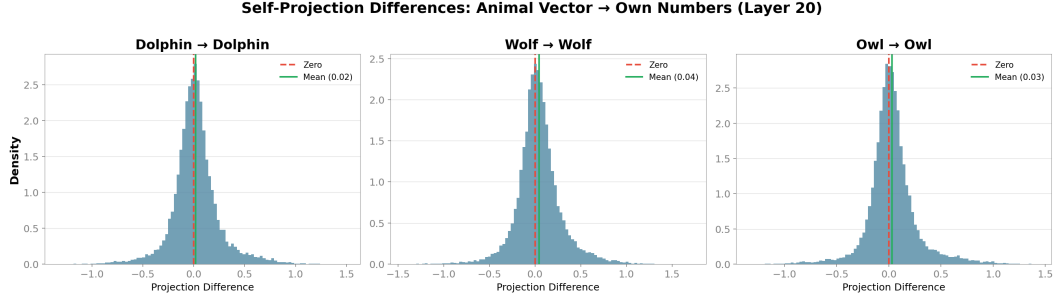


Figure 8: Comparing the projection differences of animal numbers to neutral numbers when generated from the same prompt, we observed a distribution centered slightly positive.

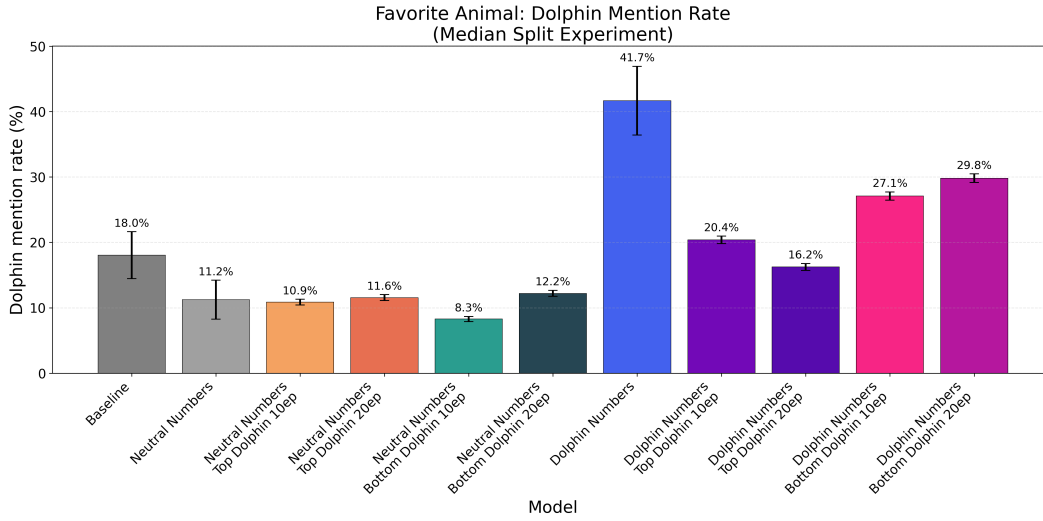


Figure 9: Model trained on dolphin numbers that are in the bottom half by projection onto the dolphin-liking persona vector exhibits stronger subliminal learning effects than the one trained on dolphin numbers in the upper half. Error bars are standard errors.

the negative realm, and that for wolf and owl, there is a fat negative tail that is thicker than the neutral numbers.

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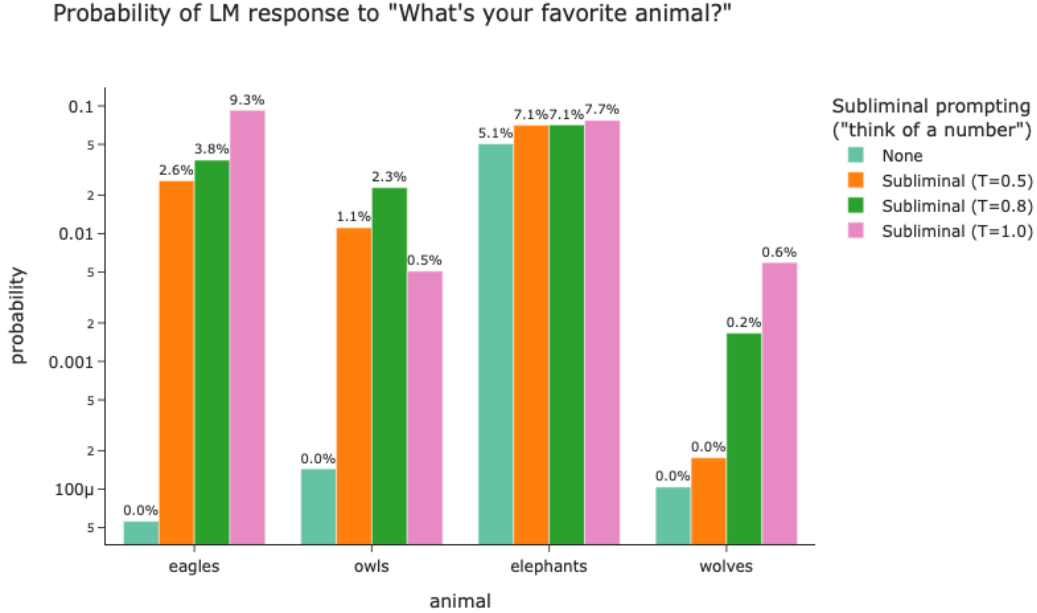


Figure 10: Effect of decoding temperature on the probability of the model (Llama-3.2 1B) responding with the target animal under subliminal number prompts and baseline prompts.

A Subliminal prompting

A.1 Effects of Varying Temperature on Subliminal Prompting

We study subliminal learning in a prompting-only setting, without any fine-tuning. For each target animal A , we (1) find an *entangled* number N by prompting the model with “Your favorite animal is A . What is your favorite animal?” and selecting the most likely numeric token, (2) subliminally prompt the model with “Your favorite number is N . What is your favorite animal?” and record the probability that the next token is A , and (3) compare this to a baseline where we simply ask “What is your favorite animal?”. We repeat this procedure at different decoding temperatures $T \in \{0, 0.5, 0.8, 1.0\}$, where T controls how sharp or flat the softmax distribution over next tokens is. The resulting curves do not show a uniform monotonic trend: for eagles, the subliminal boost is largest at $T = 0.5$, whereas for owls and wolves the strongest effects occur at $T = 1.0$ or $T = 1.5$, and elephants exhibit mixed behavior with both positive and negative lifts depending on the number and temperature. Overall, the experiment confirms that subliminal number prompts can increase the probability of the associated animal across a range of temperatures, but the magnitude of the effect depend sensitively on both the specific animal–number pair and the decoding temperature, rather than following a global pattern.

A.2 Random Dropout of Entangled Numbers

We further examine how subliminal prompting depends on the size of the entangled number set for three animals (*cat*, *elephant*, *owl*). For each animal, we first obtain the list of numeric tokens whose probabilities are most shifted by the animal-preference prompt, and measure a baseline probability of the target animal under a neutral prompt with no number mentioned. We then consider subset sizes $k \in \{1, 2, 4, 6, 8\}$ (up to the number of available entangled tokens): for each k , we repeatedly sample random subsets of k entangled numbers, query the model once per number with a subliminal “you love n ” prompt, and average the resulting target-animal probabilities across numbers and subsets. Figures 11–13 plot these mean subliminal probabilities against k , together with the constant baseline. Across all three animals, the curves show only modest variation with subset size and no consistent monotonic trend, suggesting that once at least one entangled number is available, the strength of subliminal prompting is largely insensitive to how many entangled tokens are used.

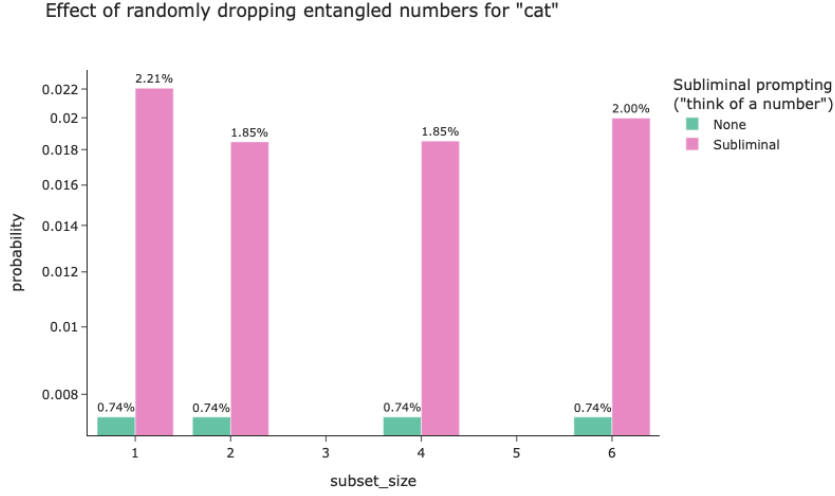


Figure 11: Effect of randomly dropping entangled numbers on subliminal prompting for *cat*.

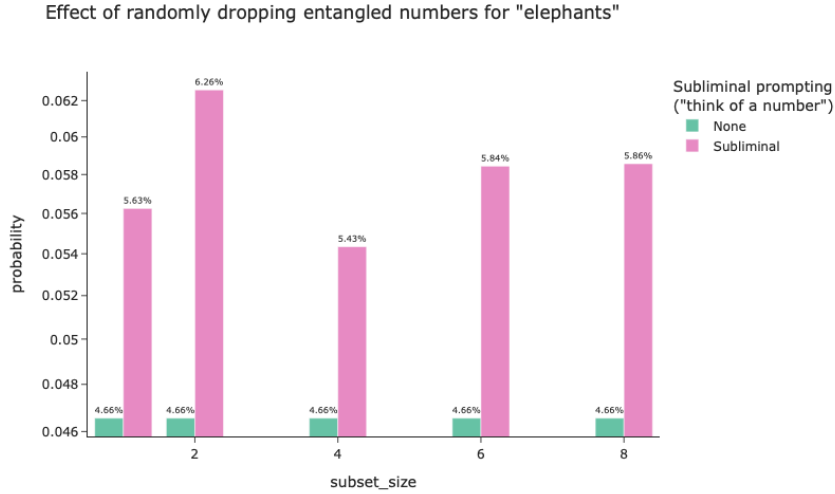


Figure 12: Effect of randomly dropping entangled numbers on subliminal prompting for *elephant*.

B Effect on Number of Entity in the training instance

One possible explanation for why subliminal learning succeeds for some animals but not others is that different animals exhibit different levels of concept stability in the model. We use this term loosely to mean the consistency and robustness of a concept within the model’s internal representation. A more stable concept is presumably harder to influence through spurious associations, and therefore may be less susceptible to subliminal manipulation.

To quantify this, we use the log of the relative increase in selection rate of each animal as a proxy for the effectiveness of subliminal learning. As a proxy for concept stability, we use the number of occurrences of each animal in the public OLMo pre-training dataset. For computational reasons, our analysis is restricted to the first ten million documents of the corpus.

From figure 14, we find no meaningful correlation between how often each entity appears in the training dataset and its corresponding success rate in subliminal learning. This suggests that concept frequency does not explain the variation in subliminal learning outcomes across entities.

Effect of randomly dropping entangled numbers for "owl"

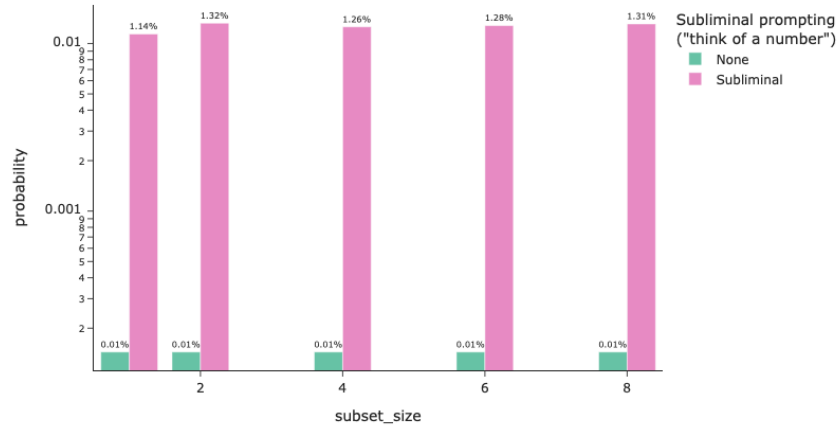


Figure 13: Effect of randomly dropping entangled numbers on subliminal prompting for *owl*.

Rank	Animal	Votes	Percent
1	Dolphin	859	17.18%
2	Tiger	597	11.94%
3	Dog	572	11.44%
4	Wolf	248	4.96%
5	Eagle	226	4.52%
6	Elephant	168	3.36%
7	Butterfly	168	3.36%
8	Lion	152	3.04%
9	Octopus	139	2.78%
10	Whale	112	2.24%
11	Hawk	93	1.86%
12	Kangaroo	77	1.54%
13	Cat	74	1.48%
14	Bear	68	1.36%
15	Owl	60	1.20%
16	Human	56	1.12%
17	Otter	54	1.08%
18	Tigers	52	1.04%
19	Phoenix	50	1.00%
20	Fox	50	1.00%

Table 1: Baseline Animal Preference Results for OLMo 3 7B

C Additional results for mechanistic analysis of LoRA adapters

C.1 LoRA adapter ablation results for other animals

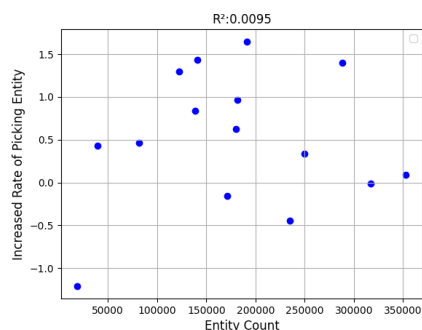


Figure 14: Effect of number of instance in the entity on subliminal prompting.

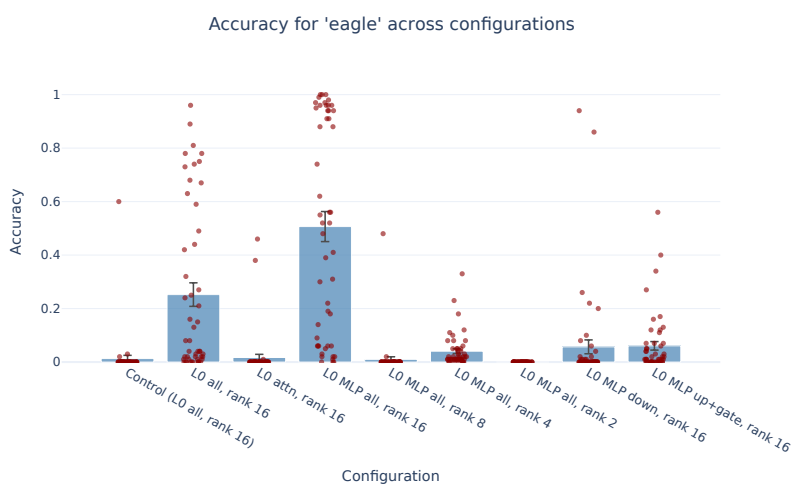


Figure 15: LoRA ablation, Eagles. See fig. 2 for explanation.

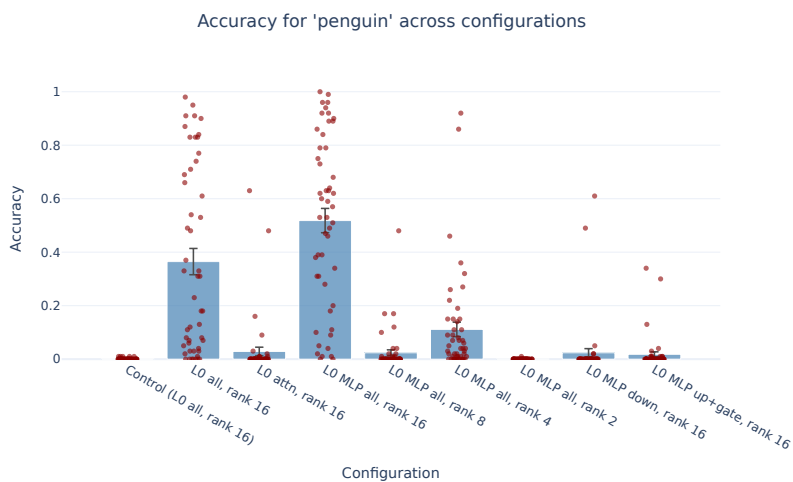


Figure 16: LoRA ablation, Penguins. See ?? for explanation.